

Database Triggers (Self-Learning)

Q. What is a Trigger in SQL. Discuss the different types of triggers (BEFORE and AFTER) and explain their primary uses.

[DEF] Definition to Remember

A Trigger is a special stored procedure in a database that automatically fires (executes) in response to a DML event (INSERT, UPDATE, or DELETE) on a specific table. Unlike regular procedures, triggers are never called explicitly -- the RDBMS engine activates them automatically.

1. Types of Triggers

Triggers are classified by when they execute relative to the triggering DML event:

A. BEFORE Trigger

Example: Before inserting into the Employee table, a BEFORE trigger checks if the incoming salary is below minimum wage. If so, it raises an error and aborts the insert.

B. AFTER Trigger

Example: After an update to Employee, an AFTER trigger inserts a record into Employee_Audit logging the user, timestamp, and what changed.

(Some systems also support INSTEAD OF Triggers -- primarily used on Views to enable update operations on non-updatable views.)

2. Trigger Event Classification

	BEFORE	AFTER
Timing	Before data is saved	After data is saved
Purpose	Validate / modify input	React / audit / cascade
**Can Abort **	Yes -- can raise error to stop	No -- data already committed

3. Uses and Advantages

Use Case	Explanation
Business Rule Enforcement	Complex rules that `CHECK` constraints
Auditing	Auto-log who changed what and when
Task Automation	Calculate derived columns, update bala
Referential Integrity	Enforce complex FK rules beyond standa

[MUST-WRITE] Must-Write Points (for 10 marks)

1. A Trigger automatically fires on a DML event (INSERT, UPDATE, DELETE) -- never called explicitly.
2. BEFORE Trigger: fires before the DML; used for data validation and input modification.
3. AFTER Trigger: fires after the DML; used for auditing and cascading operations.
4. BEFORE triggers can abort the operation by raising an error; AFTER triggers cannot.
5. Key uses: business rule enforcement, audit logging, task automation, referential integrity.
6. INSTEAD OF triggers are used on Views to allow updates on non-updatable views.

7. Poorly designed triggers can cause cascading errors and severe performance degradation.

[RECALL] Quick Recall

Trigger -> Auto-fires on INSERT/UPDATE/DELETE -> BEFORE (validate/abort) -> AFTER (audit/cascade) -> Uses: Business Rules, Auditing, Automation -> Never explicitly called